

Week 10

Communicating your findings

Outline

1. *Types and formats of communication*
2. *Elements of data visualization*
3. *Tips and tricks*

1. Types and formats of communication

How might you be asked to share your analysis?



	Presentation	Report
Purpose	Engage audience and convey specific message	Provide detailed account of information and findings
Format	Verbal with visual aids	Document (written/visual)
Structure	Can vary depending on message, employs storytelling techniques	Usually follow standardized structure
Detail	Include minimal text and data to make specific point	Include comprehensive account of data, methods, and conclusions

Example structure of a business report

PROJECT STATUS REPORT Horizon Mobile Project • HBM-015 Jennifer Mendoza																															
DECEMBER 2022 The project itself is still moving forward without any major issues. However, there were problems testing major features of the new mobile application. The VentureX feature was also not responding as expected. It is currently undergoing analysis and further testing to isolate the roots to the issues. While the project has potentially been set back by a week or two, the developers aim to resolve the outstanding issues prior to February 15, which will return the project to its initial schedule. Project Overview <table><tr><th>COMPONENT</th><th>STATUS</th><th>OWNER/TEAM</th><th>NOTES</th></tr><tr><td>Budget</td><td>Over budget</td><td>Anwar Farrid, CFO</td><td>Developers claiming overtime to deal with new issues with mobile application features.</td></tr><tr><td>Schedule</td><td>Acceptable</td><td>Elaine Krassik, Regional PM</td><td>The project schedule has not yet been compromised.</td></tr><tr><td>Quality</td><td>Less than satisfactory</td><td>Nigil Scholle, QAO</td><td>Due to early stage application bugs, the project may take longer than expected.</td></tr><tr><td>Scope</td><td>Progress paused</td><td>Tom Grayson</td><td>The overall scope of work may be reduced due to recent complications</td></tr><tr><td>Risks</td><td>Driving up project budget, releasing finished product late</td><td>Jibar Saravapudi, Risk Advisor</td><td>Project budget has already increased.</td></tr><tr><td>Roadblocks</td><td>Elusive mobile application errors</td><td>Jenny Lee, CTO</td><td>Currently running tests to isolate issues.</td></tr></table>				COMPONENT	STATUS	OWNER/TEAM	NOTES	Budget	Over budget	Anwar Farrid, CFO	Developers claiming overtime to deal with new issues with mobile application features.	Schedule	Acceptable	Elaine Krassik, Regional PM	The project schedule has not yet been compromised.	Quality	Less than satisfactory	Nigil Scholle, QAO	Due to early stage application bugs, the project may take longer than expected.	Scope	Progress paused	Tom Grayson	The overall scope of work may be reduced due to recent complications	Risks	Driving up project budget, releasing finished product late	Jibar Saravapudi, Risk Advisor	Project budget has already increased.	Roadblocks	Elusive mobile application errors	Jenny Lee, CTO	Currently running tests to isolate issues.
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1. Executive summary: concise take-home message of whole report
2. Introduction: context and motivation
3. Main body: results with supporting evidence
 - a. Can be composed of sections for each conclusion
 - b. Include data visualizations or figures to support claims
4. Conclusion: recommendations, next steps
5. Appendices: additional data, detailed methods

Example of a business presentation



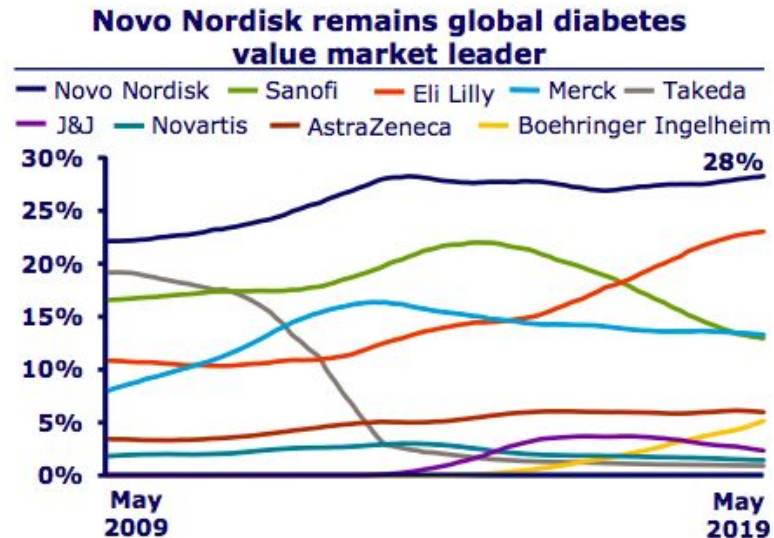
1. Engage audience by setting up the challenge or sharing interesting finding
2. Agenda or outline
3. Main content slides
 - a. Use minimal text and strong visuals
4. Key takeaways or recommendations
5. Conclusion referring back to introduction
6. Appendix slides to refer to in case of questions

2. Elements of data visualization

In presentations and reports, data visualization is key

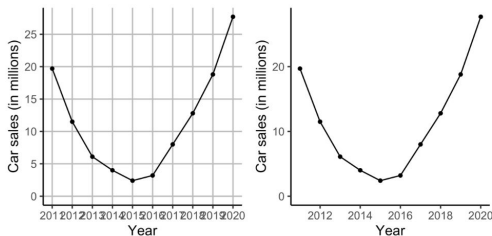
The charts and graphs you use to *analyze* your data might not be the best ones to *communicate* your results!

Data visualization in presentations should directly support your claims:

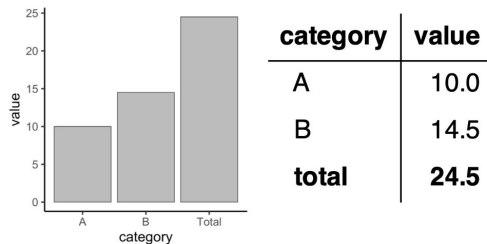


Ensure graphs are clear, readable, and not misleading:

Data-to-ink ratio



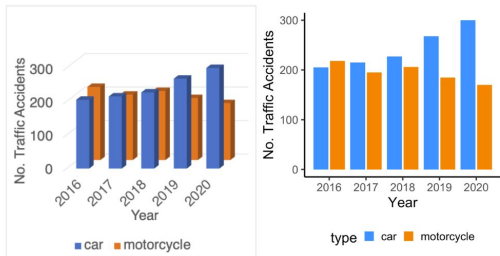
Data density



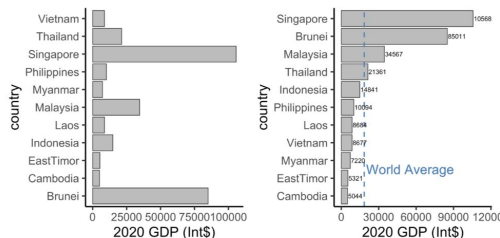
When presenting a graph, tell the audience:

- What does each axis represent?
- What does each line, point, or color represent?
- What is the takeaway from the data?

Avoid optical illusions

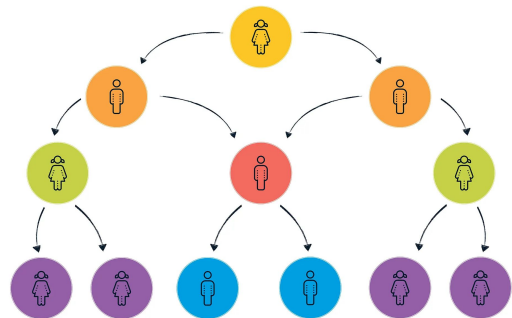


Annotate and emphasize

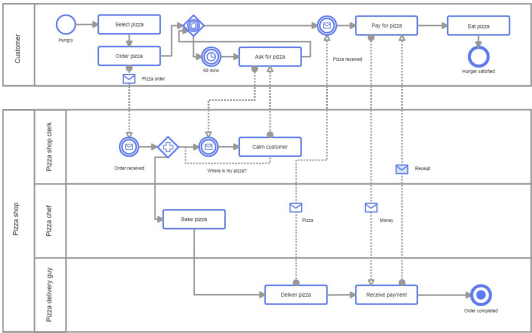


Charts are a powerful way to convey more than just numbers

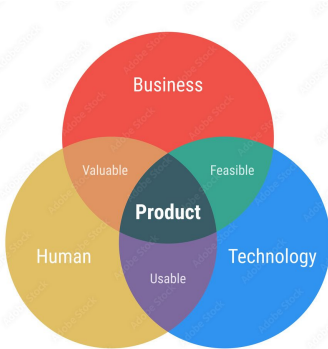
Organization structure



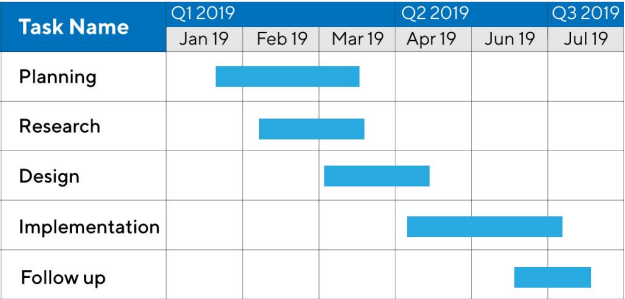
Process or workflow



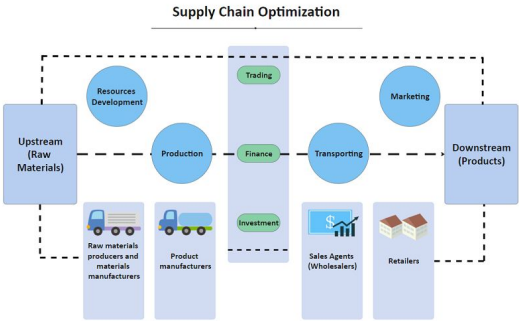
Venn diagrams



Gantt chart (timeline)



Supply chain



3. Tips and tricks

Tips and tricks for communicating your analysis

- The order you did things usually not the best order to tell things!
- Tailor your approach to your audience
- Provide context for numbers: how do they compare to an average or known reference?
- Avoid using acronyms in text or speech
- Be prepared for questions at any point during a presentation

Tools for slides: Microsoft Powerpoint, Keynote, Google Slides

Tools for documents: Microsoft Word, Google Docs, Adobe InDesign, Canva